

Person 2 — Document AI / OCR / HTR

Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform |
Step-by-step implementation specification | Unified Master Blueprint v2.0

Assigned role: Person 2 — Document AI / OCR / HTR

Document set: Unified Master Blueprint v2.0

Scope: SIH 12 + SIH 18 combined platform

Canonical rule: This document must not introduce a technology, endpoint, role, state, table, or handoff that conflicts with the Master Blueprint.

1. Mission

Own the document intelligence pipeline: preprocessing, OCR/HTR, layout analysis, structured field extraction, confidence and provenance. Deliver canonical structured outputs to the validation and orchestration layers.

2. Non-negotiable shared rules

- Use the canonical stack, API paths, entities, statuses, application roles and event names from Master Blueprint v2.0.
- Do not change another person's contract by silently renaming fields or endpoints.
- All work is project-scoped through project_id.
- All long-running work is asynchronous with job/workflow status.
- Original documents/source imagery remain immutable.
- Every AI output carries confidence, provenance, model_version and timestamp.
- Every critical mutation is auditable.
- Before handoff, run tests and provide evidence.

3. Exact handoff principle

A handoff is complete only when the receiving person can consume the documented output without guessing field names, endpoint names, status values, storage locations, or semantics.

4. Inputs

Input	Required handling
PDF/image	Validate type, size, checksum and project ownership; preserve original
Scanned pages	Render without modifying source
Handwritten registers/annotations	Route through benchmarked HTR capability where supported
Multilingual records	Use benchmark-selected language/model path; record language
Maps/cadastral documents	Preserve page image and evidence coordinates; structured extraction only where supported

5. Step-by-step pipeline

- Register the source document and immutable storage reference.
- Compute/check checksum and record metadata.
- Render PDF pages to images when necessary without replacing the original.
- Detect orientation and rotate processed copies.
- Deskew text lines.
- Denoise and improve contrast while preserving a reference to the preprocessing profile/version.
- Detect/crop document regions when useful.
- Build a benchmark containing clean printed, faded, handwritten, mixed-language, tabular and damaged samples.
- Measure character/word accuracy where labels exist and record failure cases.
- Select OCR/HTR engine/model per language and document type from benchmark evidence.
- Run OCR/HTR and retain text plus bounding boxes.
- Run layout analysis for tables, field-value regions, stamps, signatures and sections.
- Extract canonical fields: survey_number, khasra_number, khata_number, owner_details, plot_area, village, tehsil, district, land_classification, mutation_records, registration_information.

- Normalize identifiers, units and dates deterministically while preserving original_value.
- Calculate field confidence from OCR, layout, extraction and normalization signals; do not equate OCR confidence with final record confidence.
- Attach provenance: document_id, page, bbox, model_version, processed_at.
- Persist outputs and emit document.ocr.completed with the canonical payload.
- Expose failed/low-confidence outputs for validation and review; never silently drop uncertain fields.

6. Canonical field object

```
{
  "field_name": "survey_number",
  "original_value": "145/2",
  "normalized_value": "145/2",
  "confidence": 0.98,
  "source": {
    "document_id": "D-001",
    "page": 1,
    "bbox": [100, 120, 260, 160]
  },
  "model_version": "ocr-model-v1",
  "processed_at": "2026-09-11T00:00:00Z"
}
```

7. Error handling

- Unsupported MIME/content → reject before AI processing.
- Corrupt PDF/page → create explicit processing error and preserve original.
- OCR engine failure → mark stage failed; allow retry.
- Low confidence → preserve output and route to validation/review.
- Unsupported language/document type → record limitation; do not claim successful extraction.
- Model change → create a new model_version and processing version; never overwrite historical results.

8. Testing matrix

Test	Evidence
Clean printed	Expected fields and evidence boxes
Faded/skewed	Preprocessing improves or preserves measurable quality
Handwritten	HTR output with confidence and failure cases
Indian-language sample	Language/model recorded; structured extraction tested
Mixed language	Correct routing or explicit uncertainty
Tabular record	Layout regions and field extraction
Damaged page	Graceful degradation and review routing
Provenance	Every field has page/bbox/model/timestamp
Versioning	Reprocess creates new version without deleting prior output

9. Handoff to P3/P1/P6

- P3 receives structured fields, original values, normalized values, confidence, provenance and document IDs.
- P1 receives the exact document.ocr.completed event and result URI contract.
- P6 receives evidence boxes and field objects for the review workspace.

10. Definition of done

- Benchmark exists and model choice is documented.
- OCR/HTR output is structured and spatially grounded.
- At least one multilingual sample is demonstrated.
- Confidence/provenance is complete.
- Reprocessing is versioned.
- Event contract tests pass.
- No original source is overwritten.